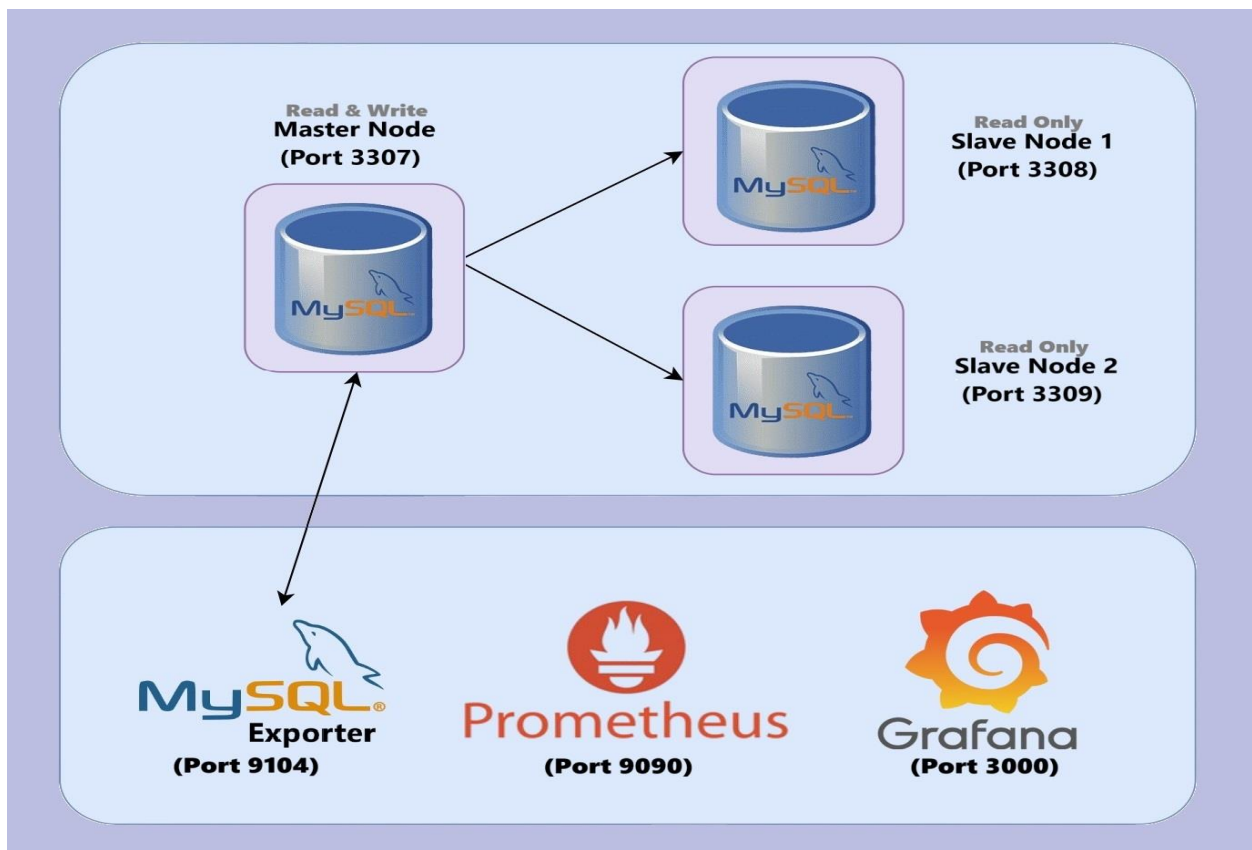


Technical Documentation for MySQL Master-Slave Replication

Overview

Master-slave database replication is a technique used to synchronize multiple copies of a database across different servers. This approach offers several benefits, primarily for load balancing and disaster recovery. By setting up a master node and slave nodes within the same MySQL cluster, we can enhance performance by directing all read operations to the slave nodes and all write operations to the master node. Additionally, this setup can serve as a disaster recovery system (DRS), providing data recovery in the event of unforeseen issues.



I will break it down to three segments in this project:

- A. Setting up a MySQL cluster with 1 master node and 2 slave nodes
- B. Setting up Grafana and Prometheus for performance monitoring
- C. Performing load testing using MySQL's built-in tool, "mysqlslap"

System and Software Requirements

1. Operating System:
 - Windows OS
2. MySQL Server:
 - MySQL 8.0
 - Cluster Configuration:
 - a. Master Server: Port 3307
 - b. Slave1 Server: Port 3308
 - c. Slave2 Server: Port 3309
3. Performance Monitoring:
 - Prometheus
 - MySQL Exporter
 - Grafana

A. Setting up a MySQL cluster with 1 master node and 2 slave nodes

1. Install MySQL 8.0 locally.
2. Create a new folder for your cluster (Ex: C:\MySQL) and subfolders for Master, Slave1 and Slave2
3. Copy the data folder from (C:\Program Files\MySQL\MySQL Server 8.0\data) to all three instances' subfolders (Ex: C:\MySQL\Master\data)
4. Remove "auto.cnf" file from all three instances' \data folder. This make sures that each instance has their own UUID. The file will be automatically created once we start MySQL.
5. Create configuration file "my.ini" in all three subfolders specifying the server_id, log-bin, port number, basedir, datadir etc. Review my.ini file for reference.

```
[mysqld]
server_id=3
log-bin=master-bin
binlog-format=ROW
# Ignore the 'mysql' database from being backed up
binlog-ignore-db=mysql

# Set to skip password authentication
# skip-grant-tables
# Set the port to 3306
port = 3307
# Set the MySQL installation directory
basedir=C:\\MySQL\\master
# Set the directory for storing MySQL database data, MySQL 8+ does not require this configuration as the sy
datadir=C:\\MySQL\\master\\data
# Allow maximum number of connections
max_connections=20
# Set the character set used by the server to UTF-8 and avoid using Latin1 encoding
character-set-server=utf8
# Set the default storage engine to be used when creating tables
default-storage-engine=INNODB

sql_mode=STRICT_TRANS_TABLES,NO_ZERO_IN_DATE,NO_ZERO_DATE,ERROR_FOR_DIVISION_BY_ZERO,NO_ENGINE_SUBSTITUTION
```

6. Install services

- Master (Port 3307)
"C:\Program Files\MySQL\MySQL Server 8.0\bin\mysqld.exe" --install MySQLMaster --defaults-file="C:\MySQL\master\my.ini"
- Slave 1 (Port 3308)
"C:\Program Files\MySQL\MySQL Server 8.0\bin\mysqld.exe" --install MySQLSlave1 --defaults-file="C:\MySQL\slave1\my.ini"
- Slave 2 (Port 3309)
"C:\Program Files\MySQL\MySQL Server 8.0\bin\mysqld.exe" --install MySQLSlave2 --defaults-file="C:\MySQL\slave2\my.ini"

To remove services if messed up "C:\Program Files\MySQL\MySQL Server 8.0\bin\mysqld.exe" --remove MySQLMaster

7. "Net start MySQLMaster" in command prompt (Administrator level).

- Logging in master instance: mysql -uroot -P3307 -p1234
- Enter the following commands in MySQL:
- CREATE USER 'sgohp'@'%' IDENTIFIED BY '1234';
- GRANT REPLICATION SLAVE ON *.* TO 'sgohp'@'%';
- FLUSH PRIVILEGES;
- Show master status;

8. "Net start MySQLSlave1" in command prompt.

- Logging in Slave 1 instance: mysql -uroot -P3308 -p1234
- Enter the following commands in MySQL:
- CREATE USER 'sgohp'@'%' IDENTIFIED BY '1234';
- *# Note that the binary log position is specified in the master status in Master instance.*
CHANGE MASTER TO MASTER_HOST='localhost', MASTER_PORT=3307,
MASTER_USER='sgohp', MASTER_PASSWORD='1234',
MASTER_LOG_FILE='master-bin.000002', MASTER_LOG_POS=1346;
- FLUSH PRIVILEGES;
- START SLAVE;
- SHOW SLAVE STATUS\G;
- Slave_IO_Running and Slave_SQL_Running should be displaying 'Yes'

```
mysql> SHOW SLAVE STATUS\G;
***** 1. row *****
Slave_IO_State: Waiting for source to send event
Master_Host: localhost
Master_User: sgohp
Master_Port: 3307
Connect_Retry: 60
Master_Log_File: master-bin.000002
Read_Master_Log_Pos: 12451538
Relay_Log_File: DESKTOP-AB8VHS5-relay-bin.000002
Relay_Log_Pos: 12450519
Relay_Master_Log_File: master-bin.000002
Slave_IO_Running: Yes
Slave_SQL_Running: Yes
```

9. "Net start MySQLSlave2" in command prompt.
 - Logging in Slave 2 instance: `mysql -uroot -P3309 -p1234`
 - Follow the rest of the steps mentioned in Step 8.
10. The setup is completed! Try creating databases or tables in Master instance to see if the replication is working.

B. Setting up Grafana and Prometheus for performance monitoring

1. Logging in Master instance: `mysql -uroot -P3307 -p1234`
2. Run the following commands to create database and grant access:
 - `CREATE DATABASE DEV_DB;`
 - `GRANT SELECT ON DB.* TO 'sgohp'@'%';`
 - `GRANT REPLICATION CLIENT,SUPER ON *.* TO 'sgohp'@'%';`
 - `GRANT ALL PRIVILEGES ON DEV_DB.* TO 'sgohp'@'%';`
 - `FLUSH PRIVILEGES;`
 - `SHOW GRANTS FOR 'sgohp'@'%';`
3. Download MySQL Exporter
 - Create my.cnf file in MySQL Exporter folder.

```
[client]
host=localhost
port=3307
user=sgohp
password=1234
```

- Run the following command line:
"C:\Program Files\prometheus-2.54.1.windows-amd64\prometheus.exe" --
config.file="C:\Program Files\prometheus-2.54.1.windows-
amd64\prometheus.yml" --web.listen-address=:9090
- MySQL Exporter should be up and running now, check this server url in
browser:
<http://localhost:9104/metrics>

MySQLd Exporter

Prometheus Exporter for MySQL servers

Version: (version=0.15.1, branch=HEAD, revision=cc349684494b5038ec5a52233bdca9eb9291e6f2)

- [Metrics](#)

4. Download Prometheus

- Add additional details in "prometheus.yml" found in the folder.

```
scrape_configs:  
  # The job name is added as a label `job`  
  - job_name: "prometheus"  
    static_configs:  
      - targets: ["localhost:9090"]  
  
  - job_name: 'mysql'  
    static_configs:  
      - targets: ['localhost:9104']
```

- Run the following command line:
→ "C:\Program Files\prometheus-2.54.1.windows-amd64\prometheus.exe" --
config.file=prometheus.yml --web.listen-address=:9090
→ Prometheus should be up and running now, check this server url in browser:
<http://localhost:9090/>

Targets

All scrape pools

AllUnhealthyCollapse All

Q

Filter by endpoint or labels

mysql (1/1 up)

show less

Endpoint	State	Labels	Last Scrape	Scrape Duration	Er
http://localhost:9104/metrics	UP	instance="localhost:9104" job="mysql" ▼	10.25s ago	35.068ms	

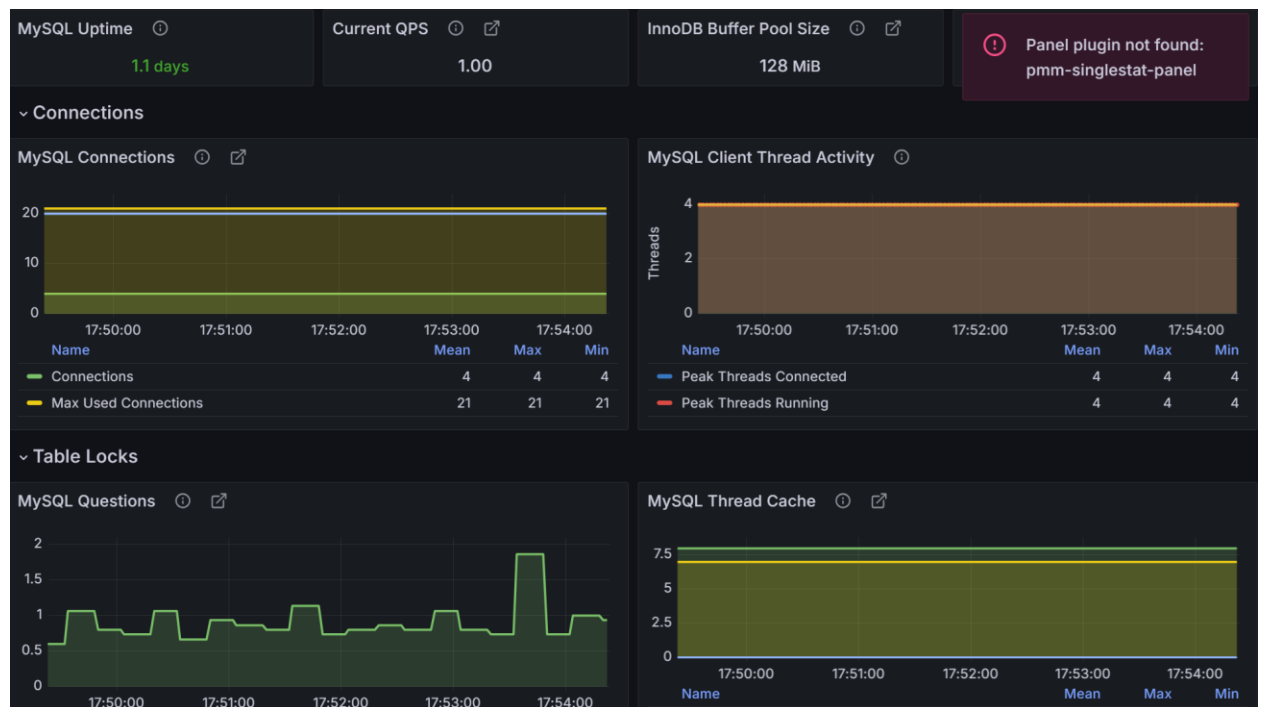
prometheus (1/1 up)

show less

Endpoint	State	Labels	Last Scrape	Scrape Duration	Er
http://localhost:9090/metrics	UP	instance="localhost:9090" job="prometheus" ▼	10.980s ago	3.006ms	

5. Download Grafana

- Prompt to server url: <http://localhost:3000>
- Create user and password. I have default 'admin' for both username and password.
- Add "Prometheus" data source with server url <http://localhost:9090>
- Import dashboard and add load MySQL Overview Dashboard with this url "<https://grafana.com/grafana/dashboards/9623-mysql-overview/>"
Grafana error log is stored here if needed: C:\Program Files\GrafanaLabs\grafana\data\log\grafana.log



C. Performing load testing using MySQL's built-in tool, "mysqlslap"

1. Logging in Master instance: `mysql -uroot -P3307 -p1234`
2. Create table in database DEV_DB for load test.
3. Run the following query:
 - `CREATE TABLE employees (
id INT AUTO_INCREMENT PRIMARY KEY,
name VARCHAR(100),
position VARCHAR(100),
salary DECIMAL(10, 2));`
 - `INSERT INTO employees (name, position, salary) VALUES
('John Doe', 'Developer', 60000.00),
('Jane Smith', 'Designer', 55000.00),
('Mike Johnson', 'Manager', 75000.00);`
4. Here are the three load tests, run this in command line:
 - **SELECT STATEMENT (Read)**
`mysqlslap --concurrency=50 --iterations=100 --query="SELECT * FROM employees" --create-schema=dev_db --user=sgohp --password=1234 --port=3307 --user=sgohp --password=1234 --port=3307`
 - **INSERT STATEMENT (Write)**
`mysqlslap --concurrency=50 --iterations=100 --query="INSERT INTO employees (name, position, salary) VALUES ('Test User', 'Tester', 50000.00);" --create-schema=dev_db --user=sgohp --password=1234 --port=3307 --user=sgohp --password=1234 --port=3307`
 - **INSERT, UPDATE, DELETE STATEMENT (Write and Read)**
`mysqlslap --concurrency=50 --iterations=100 --query="INSERT INTO employees (name, position, salary) VALUES ('Test User', 'Tester', 50000.00); UPDATE employees SET salary = salary + 5000 WHERE position = 'Tester'; DELETE FROM employees WHERE name = 'Test User';" --create-schema=dev_db --user=sgohp --password=1234 --port=3307`

```
Benchmark
Average number of seconds to run all queries: 1.056 seconds
Minimum number of seconds to run all queries: 0.281 seconds
Maximum number of seconds to run all queries: 1.703 seconds
Number of clients running queries: 50
Average number of queries per client: 1
```

Quick Notes after rebooting

- net start MySQLMaster
- net start MySQLSlave1
- net start MySQLSlave2
- "C:\Program Files\mysqld_exporter-0.15.1.windows-386\mysqld_exporter.exe" --config.my-cnf="C:\Program Files\mysqld_exporter-0.15.1.windows-386\my.cnf"
- "C:\Program Files\prometheus-2.54.1.windows-amd64\prometheus.exe" --config.file="C:\Program Files\prometheus-2.54.1.windows-amd64\prometheus.yml" --web.listen-address=:9090
- mysql -uroot -P3307 -p1234 (Master)
- mysql -uroot -P3308 -p1234 (Slave 1)
- mysql -uroot -P3309 -p1234 (Slave 2)
- <http://localhost:9104/metrics> (MySQL Exporter)
- <http://localhost:9090/> (Prometheus)
- <http://localhost:3000> (Grafana)